

Yusuf Berk AKÇAY

Curriculum Vitae

Istanbul, Turkey — yusuf.akcay1@std.bogazici.edu.tr — <https://yusufberkakcay52.github.io/>

EDUCATION

Mathematics Major with Double Major at Physics

Current Year: 3rd year

Institution: Boğaziçi University, Istanbul, Turkey

Current GPA: 3.98/4

RESEARCH INTEREST

I have a broad interest in modern aspects of theoretical and mathematical physics, especially in statistical mechanics, differential geometric formulations of classical and quantum mechanics, and dynamical systems. My Github page provides further details on my academic interests and work.

INDEPENDENT STUDIES AND ACADEMIC PROJECTS

PHYS 491: Introduction to Research in Physics I

September-December 2024

Supervisor: Asst. Prof. Ilmar Gahramanov

- Studied the Dunkl deformed quantum mechanics with an emphasis on the deformed Pauli oscillator.
- Learned about the Lewis-Riesenfeld Invariant and its uses to obtain exact solutions of the Schrödinger equation.
- Delivered a presentation on the topic to an undergraduate level audience.
- Investigated the possible supersymmetric extensions of the topic.

Course Project, PHYS 177: Computational Introduction to Dynamical Systems

July 2025

- Prepared a report on solvable structures and their relation to the Arnold-Liouville theorem.

Reading Program on Geometry of Quantization

September 2025 - Now

Supervisor: Asst. Prof. Umut Varolgüneş

- Learned about symplectic manifolds, Lagrangian submanifolds, Maslov index and related topics.
- Explored the symplectic geometric formulation of the classical mechanics, with a special emphasis on the Hamilton-Jacobi equation.
- Studied the WKB method to obtain approximate solutions of the Schrödinger equation, geometric interpretation of the WKB solutions and Maslov correction.
- Studied the geometric quantization of general symplectic manifolds.

SCHOOLS AND WORKSHOPS

Istanbul Stringy Meeting

Boğaziçi University, Istanbul, Turkey

2024

COSMOVERSE@ISTANBUL 2025

Istanbul Technical University, Istanbul, Turkey

2025

Grad. and Undergrad. Summer Math School

Nesin Mathematics Village, Izmir, Turkey

2025

- Category Theory
- Conformal Field Theory
- Selected Topics in Modern Physics

37th National Mathematics Symposium

Akdeniz University, Antalya, Turkey

2025

Summer School on the Digital Archive of Exact Solutions in General Relativity

Istanbul Technical University, Istanbul

2025

- Learned fundamental principles of general relativity, aiming to understand the Newman-Penrose formalism, Killing equation and its solutions.
- Used Sage-Math to perform calculations in Newman-Penrose formalism.
- Studied the Lie algebras, transformation groups in general relativity and their classification.

Istanbul Stringy Meeting

Boğaziçi University, Istanbul, Turkey

2026

TALKS

Workshop on Symmetry and Geometry in Memory of Yavuz Nutku

Hydrodynamic Systems with Two Conserved Quantities

2026

- Integrability and Bi-Hamiltonian structures of d-Boussinesq and Benney-Lax equations

Boğaziçi Math Grad Seminars

WKB Approximation and Geometric Quantization

2026

- Connection between the WKB approximation and the framework of Geometric Quantization

Directed Reading Program 2026

N=4 Super Yang-Mills Theory

2026

- Introduction to Quantum Field Theory, with an emphasis on renormalization, Yang-Mills theory, supersymmetry, and the construction of the N=4 supersymmetric Yang-Mills Lagrangian

EXAMS

- Scored 500.00/500 on the LGS 2018 (High School Entrance Exam).
- Passed the TUBITAK Physics Olympiad 1st stage with a score of 10.25/25, thus earning the right to participate in the 2nd stage among 598 participants.
- Passed the TUBITAK Physics Olympiad 2nd stage with a score of 34.5/100, thus earning a Bronze medal and the right to participate in the 3rd stage among 56 participants.
- Admitted to Boğaziçi University with a score of 558.77668/560 on the YKS 2023 (Higher Education Institutions Examination).

SKILLS

Languages: **Turkish**(native), **English**(fluent)

Computational Tools: **Python**, **Mathematica**, **LaTeX**

SCHOLARSHIPS

TUBITAK 2205 Undergraduate Scholarship

EXTRACURRICULAR ACTIVITIES

- Licensed basketball player for 2 years.
- Licensed scout for 4 years.
- Organized voluntary lectures for high school students.